

MATH9102

COMPARING MORE THAN TWO GROUPS

Sources used in creation of this lecture:

Statistics and Data Analysis, Peck, Olsen and Devore; Discovering Statistics Using IBM SPSS, Andy Field; Understanding Basic Statistics, Brase and Brase; SPSS Survival Manual, Julie Pallant

What we have covered so far...

Theme	Located in Lectures	Core Skills
Foundations and Measurement	Weeks 1, 2	Understanding variables, data types, sampling, measurement, bias
Descriptive Statistics	Weeks 1, 2	Central tendency, dispersion, visualisation
Data Quality and Preparation	Week 3, Week 5	Handling missing data, outliers, weighting, normality; Transforming data
Inferential Statistics – Foundations	Week 3	Hypothesis testing, p-values, assumptions
Relationships	Week 3	Correlation analysis
Comparing Categorical	Week 4	
Comparing Groups	Week 4	Continuous and binary categorical
Comparing Groups	Week 5 (This week)	Continuous and multiple categorical

Comparison of more than 2 samples



Tell me the
difference
between these
groups...
Thank God I have
ANOVA

Tests for Multiple Group Comparison

	Independent Samples	Related Samples
Interval measures/ parametric	ANOVA* 	Repeated** Measures ANOVA 
Ordinal/ non-parametric	Kruskal-Wallis	Freidman

* multiple different groups of participants

** same participants measured at multiple different points

Parametric

MATH9102-W4-ANOVA.QMD

ANOVA

ANalysis Of VAriance (ANOVA)

- Still compares the differences in means between groups but it uses the variance of data to “decide” if means are different
- Really is ANOVASMD (**A**nalysis **o**f **v**ariance to see if **m**eans are **d**ifferent)

Looks to see what the variation (variance) is *within* the groups, then works out how that variation would translate into variation (i.e. differences) *between* the groups, taking into account how many subjects there are in the groups.

If the observed differences are a lot bigger than what you'd expect by chance, you have statistical significance.

ANOVA

Factors: the overall 'things' being compared (e.g. age, task, score)

Levels: the elements of the factor (young v old and naming v reading aloud)

F- statistic or F-ratio

- Magnitude of the difference between the different conditions
 - Similar to z or t-score as it compares the amount of systematic variance in the data to the amount of unsystematic variance.
- It is the ratio of the experimental effect to the individual differences in performance.
- Less than 1, it must represent a non-significant event (so you always want a F-ratio greater than 1)
- Degrees of freedom – depends on the number of factors and the number of levels

ANOVA

ANOVA tests for one overall effect only

- **Omnibus test**

There is a need for post-hoc testing

- ANOVA can tell you if there is an effect but not where
- Need to perform multiple comparisons to identify level of effect and where effect is evident

To test for significance

- obtained F-ratio is compared against maximum value one would expect to get by chance alone in an F-distribution with the same degrees of freedom.
- p-value associated with F is probability that differences between groups could occur by chance if null-hypothesis is correct

Reporting convention: $F = 65.58$, $df = 4, 45$, $p < .001$

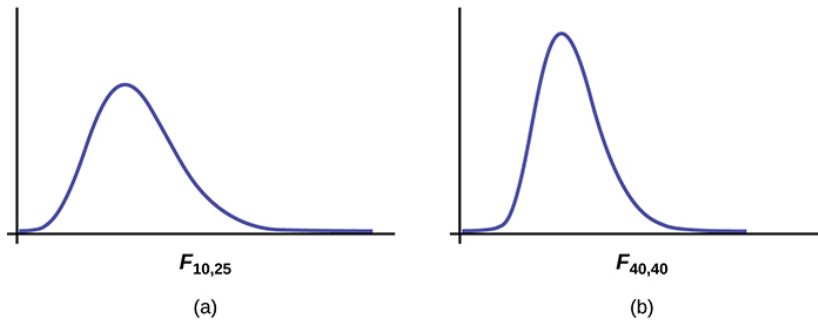
F Distribution

The curve is not symmetrical but skewed to the right.

There is a different curve for each set of degrees of freedom (*dfs*).

The *F* statistic is greater than or equal to zero.

As the degrees of freedom for the numerator and for the denominator get larger, the curve approximates the normal.



What Does ANOVA Tell us?

Null Hypothesis:

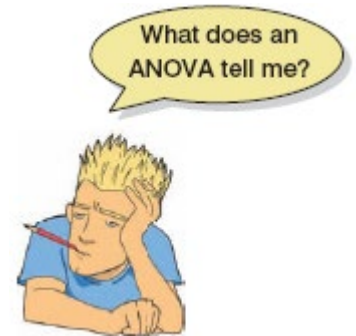
- Like a t-test, ANOVA tests the null hypothesis that the means of the different groups are the same.

Alternate Hypothesis:

- The means differ.

ANOVA is an Omnibus test

- It test for an overall difference between groups.
- It tells us that the group means are different.
- It doesn't tell us exactly which means differ.



Example 1

Sleep.rdata

- <http://spss.allenandunwin.com.s3-website-ap-southeast-2.amazonaws.com/data-files.html#.Wb0vvvWP-po>
- Created from the SPSS file sleep.sav taken from SPSS Survival Manual 6th Edition Julie Pallant

A real data file condensed from a study conducted to explore the prevalence and impact of sleep problems on various aspects of people's lives.

Staff from a university in Melbourne, Australia were invited to complete a questionnaire containing questions about their sleep behavior (e.g. hours slept per night), sleep problems (e.g. difficulty getting to sleep) and the impact that these problems have on aspects of their lives (work, driving, relationships).

The sample consisted of 271 respondents (55% female, 45% male) ranging in age from 18 to 84 years (mean=44yrs).

Example: One-way Between-Groups ANOVA (sleep.rdata)

Question

- Does age impact Total Sleepiness And Associated Sensation scores of participants?

Need

- One independent categorical (nominal/ordinal) variable with three or more levels (age category) (`agegp3`)
 - Group 1= ≤ 37 yrs , Group 2=38-50yrs, Group 3=51+yrs (older)
- One continuous (ratio) variable (Total Sleepiness and associated sensation scale scores) (`totsas`)

Hypotheses:

- H_0 : There are no differences between Total Sleepiness And Associated Sensation Scale scores for respondents in Group 1, Group2 and Group3.
- H_A : There are differences between Total Sleepiness And Associated Sensation Scale scores for respondents in Group 1, Group2 and Group3.

One-way Between-Groups ANOVA

One-way ANOVA will tell us

- whether there are significant differences in Total Sleepiness and associated sensation scale scores (`totsas`) across the three groups (`agegp3`)

BUT

- **Not where** the difference is

We need to conduct post-hoc tests (additional tests after the ANOVA) to find out where the differences lie

- E.g. Is it between young and older, young and middle-aged, middle-aged and older?

One-way Between- Groups ANOVA Step 1

Before we conduct the ANOVA we would first have inspected and described the data for that variable as we would any scale (interval/ratio) variable in advance to see if it can be considered normal.

Generate and Inspect:

- Histogram with density curve
- QQ Plot
- Standardized scores for skewness and kurtosis
- Percentage of absolute (z scores) outside acceptable ranges
- Appropriate descriptive statistics

One-way Between-Groups ANOVA

Reporting our findings:

“Total Sleepiness and associated sensation scores data was assessed to see if it followed a normal distribution. Visual inspection of the histogram and QQ-Plot (see Figure x) identified some issues with skewness and kurtosis. The standardised score for skewness (-0.37) can be considered acceptable using the criteria proposed by West, Finch and Curran (1996), however the standardised score for kurtosis (-3.65) was outside the acceptable range. However as 100% of standardised scores for total sleepiness and associated sensation fell within the bounds of ± 3.29 , the data can be considered to approximate a normal distribution as outlined in Field, Miles and Field (2013).”

West SG, Finch JF, Curran PJ. Structural equation models with nonnormal variables: problems and remedies. In: Hoyle RH, editor. Structural equation modeling: Concepts, issues and applications. Newbery Park, CA: Sage; 1995. pp. 56–75.

One Way ANOVA Step 2

Generate summary statistics for the variable of interest for each group.

These three approaches will all work:

```
by(sleep$totsas, survey$agegp3,  
summary)
```

```
psych::describeBy(survey$totsas  
, survey$agegp3)
```

```
tapply(survey$totsas,  
survey$agegp3, summary)
```


One Way ANOVA Step 2 Generate Summary Statistics per Group

Generate summary statistics for the variable of interest for each group.

```
# item group1 vars n mean sd  
median trimmed mad min max
```

```
Group 1 <= 37 n=79, m=27.77, sd=10.30
```

```
Group 2 38 - 50 n= 79, m=26.07, sd= 11.18
```

```
Group 3 51+ n=72, m=25.14, sd=9.71
```

One Way ANOVA Step 3 Testing for homogeneity of variance in R

We need to check for homoscedasticity (homogeneity of variance) for the variable of interest across groups

In R we use Bartlett's test from the library stats

```
stats::bartlett.test(sleep$tots  
as, sleep$agegp3)
```

Can be argued that the variances are homogeneous if the *p-value* > 0.05

Testing for homogeneity of variance in R

We need to check for homoscedasticity (homogeneity of variance) for the variable of interest across groups

In R we use Bartlett's test from the library stats

Can be argued that the variances are homogeneous if the *p-value* > 0.05

HERE: Bartlett's K-squared = 1.49, df = 2, p-value = 0.48

```
#Conduct Bartlett's test for homogeneity of variance in library car - the null hypothesis is that variances in groups are equal so to assume homogeneity we would expect probability to not be statistically significant.  
stats::bartlett.test(totsas~ agegp3, data=sleep)
```

```
##  
## Bartlett test of homogeneity of variances  
##  
## data:  totsas by agegp3  
## Bartlett's K-squared = 1.4867, df = 2, p-value = 0.4755
```

Testing for homogeneity of variance

We need to know so we can
conduct the correct post-hoc
test

- Homoscedasticity:
 - Use Tukey's honestly significant difference (HSD) post hoc test (tukey)
- Heterocedasticity:
 - Use Games-Howell (games-howell)

One Way ANOVA Step 4 Conduct the Omnibus test

Compute the analysis of variance using the

We can set VAR.equal to be true because we have homogeneity

- If we don't we set it to FALSE

In R – One Way ANOVA

```
# Compute the analysis of variance using the  
oneway.test from the stats package- we set VAR.equal  
to be true because we have homogeneity
```

```
res <- stats::oneway.test(sleep$totsas ~  
as.factor(sleep$agegp3), data = sleep,  
var.equal = TRUE)
```

```
# Summary of the analysis - echo to screen
```

```
res
```

In R – One Way ANOVA

```
##  
## One-way analysis of means  
##  
## data:  sleep$totsas and as.factor(sleep$agegp3)  
## F = 1.245, num df = 2, denom df = 227, p-value = 0.2899
```

Check this first!

If this is not significant you do not need to check the post hoc test results.

One Way ANOVA Step 5

If omnibus is
significant
Conduct the
Post-hoc test

- **Homoscedasticity:**
 - Use Tukey's honestly significant difference (HSD) post hoc test (tukey)
- **Heterocedasticity:**
 - Use Games-Howell (games-howell)

In R – One Way ANOVA

```
#No statistically significant difference was found, p=0.29  
#We don't need the post-hoc but because variances were equal we can use the tukey test from the rstatix package  
#This is for example only - it is not relevant for our example  
rstatix::tukey_hsd(sleep, totsas ~ agegp3)
```

```
## Warning: Expected 2 pieces. Additional pieces discarded in 2 rows [1, 3].
```

term <chr>	group1 <chr>	group2 <chr>	null.value <dbl>	estimate <dbl>	conf.low <dbl>	conf.high <dbl>	p.adj <dbl>	p.adj.signif <chr>
1 agegp3	50	38	0	-1.6962025	-5.614200	2.221795	0.564	ns
2 agegp3	<= 37	51+	0	-2.6332630	-6.645360	1.378834	0.270	ns
3 agegp3	38	51+	0	-0.9370605	-4.949157	3.075036	0.846	ns

3 rows

Here we see the differences between groups

- We would be looking for p-values < 0.05 – these are where the differences lie
- We only need to worry about this post-hoc test if the omnibus test returned a significant value

One Way ANOVA Step 6 Calculating the effect size

eta squared = sum of squares between groups / total sum of squares (from our ANOVA output (rounded up))

Guidelines on effect size:

0.01 = small

0.06 = moderate

0.14 = large

For our example it is $\approx .01$

Step 7 Report the result

A one way ANOVA test was conducted to compare Total Sleepiness and associated sensation scale score for respondents who are young (18 to 37), middle-aged (38 to 50) and older (over 50). No statistically significant difference in the scores was found between age groups, ($F(2, 2, 227) = 1.24, p=1.245$), (($M=27.77, SD=10.3$) for respondents who are aged 37 and under, ($M=26.08, SD=11.18$) for respondents who are aged between 38 and 50, $M=25.14, SD=9.71$) for respondents who are aged 50+). A small effect size was indicated by the eta squared value ($\eta^2 = .01$). Therefore, there is no evidence to reject the null hypothesis that Total Sleepiness and associated sensation scores differ by age group.

Example 2: One-way Between-Groups ANOVA Survey.dat

survey.dat (Julie Pallant)

Question

- Does age impact optimism scores of respondents?

Need

- One independent variable with three or more levels (age category) (agegp3)
- Group 1=18-29yrs, Group 2=30-44yrs, Group 3=45+yrs
- One continuous variable (optimism scores) (toptim)

Example 2: One-way Between-Groups ANOVA Survey.dat

survey.dat (Julie Pallant)

Question

- Does age impact optimism scores?

H0: There are no differences between Total Optimism scores for respondents in Group 1, Group2 and Group3.

HA: There are differences between Total Optimism scores for respondents in Group 1, Group2 and Group3.

Assess toptim for normality

Total optimism scores data were assessed for normality. Visual inspection of the histogram and QQ-Plot (see Figure 1 and Figure 2) identified some issues with skewness and kurtosis. The standardized score for kurtosis (0.91) can be considered acceptable using the criteria proposed by West, Finch and Curran (1996), but the standardized score for skewness (-4.21) was outside the acceptable range. However over 99% of standardized scores for total optimism fall within the bounds of ± 3.29 , using the guidance of Field, Miles and Field (2013) the data can be considered to approximate a normal distribution ($m=22.12$, $sd=4.43$, $n=435$).

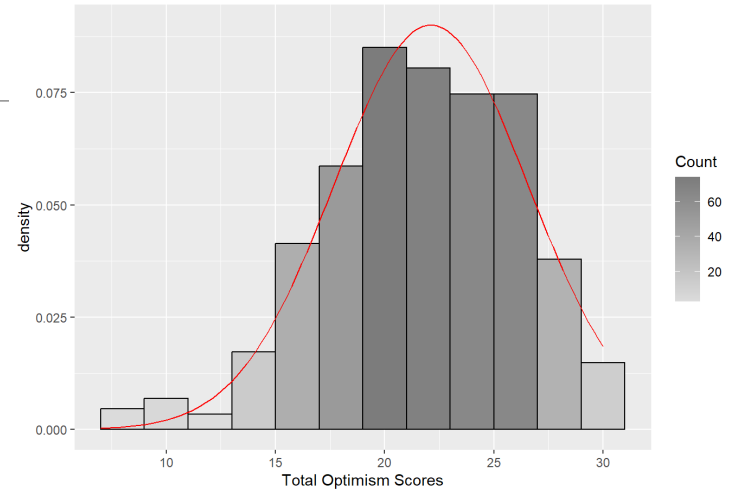
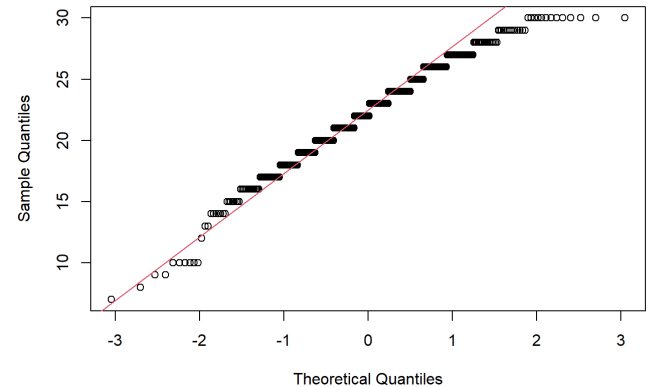


Figure 8: Histogram for Total Optimism Scores

Figure 9: QQ Plot for Total Optimism



group	n	mean	sd
1 18 - 29	147	21.36	4.55
2 30 - 44	153	22.10	4.15
3 45+	135	22.96	4.49

Central
Tendency and
Dispersion for
each group

Check for homogeneity of variance in the groups

```
stats::bartlett.test(toptim~ agegp3, data=survey)
```

```
##  
## Bartlett test of homogeneity of variances  
##  
## data:  toptim by agegp3  
## Bartlett's K-squared = 1.4561, df = 2, p-value =  
0.4828
```

The p value is > 0.05 so the result is not statistically significant so we can assume homogeneity

Conduct the ANOVA omnibus

#Conduct ANOVA using the one-way ANOVA from the stats package with var.equal=TRUE

```
stats::oneway.test(toptim ~ as.factor(agegp3), data =  
survey, var.equal = TRUE)
```

```
##
```

```
## One-way analysis of means
```

```
##
```

```
## data: toptim and as.factor(agegp3)
```

```
## F = 4.6411, num df = 2, denom df = 432, p-value = 0.01013
```

P value is < 0.05 so we have a statistically significant result

#Use Tukey

term	group1	group2	null.value	estimate	conf.low	conf.high	p.adj	p.adj.signif
<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1 agegp3	44	30	0	0.7440309	-0.4489781	1.937040	0.30800	ns
2 agegp3	18	45+	0	1.5950113	0.3636472	2.826376	0.00693	**
3 agegp3	30	45+	0	0.8509804	-0.3687705	2.070731	0.23000	ns
3 rows								

Calculate the effect size

```
effectsize :: effectsize(res)
```

	Eta2 <dbl>	CI <dbl>	CI_low <dbl>	CI_high <dbl>
F	0.02103477	0.95	0.002877522	1
1 row				

Reporting the result

A one-way between-groups analysis of variance (ANOVA) was conducted to explore the impact of age on levels of optimism, as measured by the Life orientation Test (LOT). Participants were divided into three groups according to their age (Group 1: 29 yrs or less; Group 2: 30 to 44 yrs; Group 3: 45yrs and above). There was a statistically significant difference in LOT scores for the three age groups: $F(2, 432) = 4.641$, $p < 0.05$. Despite reaching statistical significance, the actual difference in mean scores between groups was quite small. The effect size, calculated using eta squared was (0.02). Post-hoc comparisons using the Tukey HSD test indicated that the mean score for Group 1 ($M = 21.36, SD = 4.55$) was significantly different to that for Group 3 ($M = 22.96, SD = 4.49$). Group 2 ($M = 22.1, SD = 4.15$) did not differ significantly from Group 1 or Group 3. Therefore while there is evidence to reject the null hypothesis in favour of the alternate hypothesis that there are differences in optimism between age groups, this difference was found to exist only between two groups and was quite small.

Types of ANOVAs

Type	2-way ANOVA for independent groups			repeated measures ANOVA			mixed ANOVA		
Participants		Condition I	Condition II		Condition I	Condition II		Condition I	Condition II
	Task I	Participant group A	Participant group B	Task I	Participant group A	Participant group A	Task I	Participant group A	Participant group B
	Task II	Participant group C	Participant group D	Task II	Participant group A	Participant group A	Task II	Participant group A	Participant group B

Between-subject design

Within-subject design

both

NOTE: You may have more than 2 levels in each condition/ task

Non-Parametric

MATH9102-W4-KRUSKAL.QMD

Kruskal–Wallis test – Non Parametric

The Kruskal–Wallis test (Kruskal & Wallis, 1952;) is the non-parametric counterpart of the one-way independent ANOVA (analysis of variance).

The theory is very similar to that of the Mann–Whitney (and Wilcoxon rank-sum) test:

- It is based on ranked data.
- The sum of ranks for each group is denoted by R_i (where i is used to denote the particular group).

Kruskal-Wallis test Theory

Once the sum of ranks has been calculated for each group.

The test statistic, H , is calculated as:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{n_i} - 3(N+1)$$

- R_i is the sum of ranks for each group.
- N is the total sample size.
- n_i is the sample size of a particular group.

Example

SPSS file taken from
Quantitative Data
Analysis in Education,
Paul Connolly

Question:

- “Is there a significant difference in students’ GCSE mathematics grades across different family social class groups ?
- H_0 : There is no significant difference in the GCSE Maths grades achieved by across different family social class groups.
- H_A : There is no significant difference in the GCSE Maths grades achieved by across different family social class groups.
- famsec: A categorical variable representing family social class (grouped by NS-SEC).
 - 1 Large employers and higher professionals, 2 Lower professional and higher technical occupations, 3 Intermediate occupations, 4 Lower supervisory occupations, 5 Semi routine and routine occupations, 6 Other
- gradmath: grade achieved in maths (ordinal)
 - -9 Didn’t take 1 A* 2 A 3 B 4 C 5 D 6 E 7 F 8 G 9 Fail

Provisional Analysis – Statistics for gradmath grouped by socialclass

```
# Ensure gradmath is an ordered factor
ydata$gradmath <- factor(
  ydata$gradmath,
  levels = c("A*", "A", "B", "C", "D", "E", "F"),
  ordered = TRUE
)

# Summarise median and IQR
ydata %>%
  group_by(famsec) %>%
  summarise(
    `Class` = unique(famsec),
    `Median Maths Grade` = {
      num <- as.numeric(gradmath)
      med_num <- median(num, na.rm = TRUE)
      levels(gradmath)[round(med_num)]
    },
    IQR = {
      num <- as.numeric(gradmath)
      q1 <- quantile(num, 0.25, na.rm = TRUE)
      q3 <- quantile(num, 0.75, na.rm = TRUE)
      paste0(levels(gradmath)[round(q1)], "-")
    }
  ) %>%
  select(`Class`, `Median Maths Grade`, IQR)
```

Provisional
Analysis –
Statistics for
gradmath
grouped by
socialclass

Class	Median Maths Grade	IQR
<chr>	<chr>	<chr>
Intermediate occupations	C	B–D
Large employers and higher professionals	B	A–C
Lower professional and higher technical occupations	C	B–D
Lower supervisory occupations	D	C–E
Other	D	B–E
Semi routine and routine occupations	D	C–E

In R – Do the Omnibus test

```
Stats::kruskal.test(gradmath~famsec,data=ydata)
```

```
Kruskal-wallis rank sum test
```

```
data: gradmath by famsec
```

```
Kruskal-wallis chi-squared = 1556.9, df = 5, p-value < 2.2e-16
```

The test statistic (Kruskal-Wallis chi-squared) is referred to as **H**.

Posthoc – Dunn's Test Approach 1

```
#We need the as.numeric to make sure our gradmath variable is treated as a numeric variable
#and we are using the numerical categories for famsec so we use the factors
tmp <- FSA::dunnTest(as.numeric(ydata$gradmath),ydata$famsec, method = "bonferroni")

# Convert the Dunn's test result to a data frame
dunn_test_df <- as.data.frame(tmp$res)

# Filter for statistically significant comparisons (e.g.,  $p < 0.05$ )
significant_results <- dunn_test_df[dunn_test_df$P.adj < 0.05, ]

# Round the p-values to 2 decimal places
significant_results$P.adj <- round(significant_results$P.adj, 2)
# Round the p-values to 2 decimal places
significant_results$P.unadj <- round(significant_results$P.unadj, 2)

# Create a flextable to display the statistically significant results
ft <- flextable(significant_results) %>%
  set_table_properties(width = 1, layout = "autofit") # Autofit to ensure all content is visible

# Display the formatted flextable
ft
```

In R – Posthoc testing

Comparison	Z	P.unadj	P.adj
Intermediate occupations - Large employers and higher professionals	20.509856	0	0
Intermediate occupations - Lower professional and higher technical occupations	9.500534	0	0
Large employers and higher professionals - Lower professional and higher technical occupations	-12.149061	0	0
Intermediate occupations - Lower supervisory occupations	-10.502992	0	0
Large employers and higher professionals - Lower supervisory occupations	-25.448639	0	0
Lower professional and higher technical occupations - Lower supervisory occupations	-17.531140	0	0
Intermediate occupations - Other	-11.080126	0	0
Large employers and higher professionals - Other	-27.381147	0	0
Lower professional and higher technical occupations - Other	-18.844822	0	0
Intermediate occupations - Semi routine and routine occupations	-13.249197	0	0
Large employers and higher professionals - Semi routine and routine occupations	-29.718216	0	0
Lower professional and higher technical occupations - Semi routine and routine occupations	-21.182384	0	0

In R – Reporting Posthoc testing Approach 2

```
#Print the post hoc test results as a table- filtering out only the statistically significant results
tmp2 <- rstatix::dunn_test(ydata, gradmath ~ famsec, p.adjust.method = "bonferroni", detailed = FALSE) %>%
  dplyr::filter(p.adj < 0.05)
```

```
tmp2
```

.y. <chr>	group1 <chr>	group2 <chr>	n1 <int>	n2 <int>	statistic <dbl>
gradmath	Intermediate occupations	Large employers and higher professionals	2878	2736	-20.35185
gradmath	Intermediate occupations	Lower professional and higher technical occupations	2878	3532	-9.56670
gradmath	Intermediate occupations	Lower supervisory occupations	2878	1041	10.30632
gradmath	Intermediate occupations	Other	2878	1355	11.58497
gradmath	Intermediate occupations	Semi routine and routine occupations	2878	1414	14.03721
gradmath	Large employers and higher professionals	Lower professional and higher technical occupations	2736	3532	11.90458
gradmath	Large employers and higher professionals	Lower supervisory occupations	2736	1041	25.15868
gradmath	Large employers and higher professionals	Other	2736	1355	27.84865
gradmath	Large employers and higher professionals	Semi routine and routine occupations	2736	1414	30.51059
gradmath	Lower professional and higher technical occupations	Lower supervisory occupations	3532	1041	17.38147

1-10 of 12 rows | 1-6 of 9 columns

Previous 1 2 N

What is this Bonferroni Correction?

When you conduct several statistical tests simultaneously, the chance of finding at least one statistically significant result by random chance increases.

- This inflates the chance of making a Type I error (incorrectly rejecting the null hypothesis)

The Bonferroni correction addresses this by making the significance threshold more stringent.

- Bonferroni corrects/adjusts the p value by dividing the original α -value by the number of analyses on the dependent variable.
- The correction is applied by dividing the original significance level (e.g., 0.05) by the number of comparisons being made.
 - For example, if you are performing 10 tests and the original significance level is 0.05, the adjusted significance level would be $0.05/10=0.005$.
 - Each individual p-value must then be less than 0.005 to be considered statistically significant.
- It is however vulnerable to Type II error

Beware! Multiple Comparison Problem

When reading a study, you can only account for multiple comparisons when you know about all the comparisons made by the investigators.

If they report only “significant” differences, without reporting the total number of comparisons, it is not possible to properly evaluate the results.

Ideally, all analyses should be planned before collecting data, and all should be reported.

Calculate an Effect Size (eta squared)

$$\eta_H^2 = \frac{H - k + 1}{n - k}$$

- H – the value obtained in the Kruskal-Wallis test (the Kruskal-Wallis H -test statistic)
- η^2 – eta-squared estimate assumes values from 0 to 1 and multiplied by 100% indicates the percentage of variance in the dependent variable explained by the independent variable
- k – the number of groups
- n – the total number of observations

Can use an effect calculator – in R `rstatix` package contains a function `rstatix::kruskal_effsize(ydata, gradmath~famsec, ci = FALSE, conf.level = 0.95, ci.type = "perc", nboot = 1000)` #uses bootstrapping

Reporting the Results

You report in the same manner as ANOVA except

- You report that a Kruskal-Wallis test was conducted
- The omnibus test statistic is H
- Report the effect
- Report the median and IQR of those where differences applied

Reporting the Results

A Kruskal–Wallis test was conducted to examine differences in mathematics grades across family social class groups. The results indicated a statistically significant difference in median maths grades between groups, $H(5) = , p < .001$, suggesting that family social class is associated with student performance in mathematics. Post hoc pairwise comparisons using Dunn's test with Bonferroni adjustment revealed that students from higher social class backgrounds (e.g., Large employers and higher professionals) achieved significantly higher maths grades than those from lower social class groups (e.g., Semi-routine and routine occupations, Lower supervisory occupations). Differences between adjacent middle groups (e.g., Intermediate occupations vs. Lower professional and higher technical occupations) were smaller but still significant ($p < .001$). Overall, the pattern of results suggests a social gradient in mathematics achievement, where students from families with higher occupational status tend to obtain higher grades. The effect size, calculated using eta squared was moderate at 0.12. The median and IQR values for the GCSE Maths grades for all class categories are shown in the following table:

Maths Grades by Family Social Class

	Class	n	Median.Maths.Grade	IQR
Intermediate occupations	Intermediate occupations	2927	C	B– D
Large employers and higher professionals	Large employers and higher professionals	2762	B	A– C
Lower professional and higher technical occupations	Lower professional and higher technical occupations	3566	C	B– D
Lower supervisory occupations	Lower supervisory occupations	1059	D	C–E
Other	Other	1420	D	B–E
Semi routine and routine occupations	Semi routine and routine occupations	1467	D	C–E

Friedman's ANOVA

Non-parametric test for differences between several related groups

Used for testing differences between conditions when:

- There are more than two conditions
- The same participants have been used in all conditions (each case contributes several scores to the data).

If you have violated some assumption of parametric tests, then this test can be a useful way around the problem.